

## Øving 10

lørdag 22. mars 2025

17:36

### Problem 1

a)  $0 < m_a \leq 1$  and DC: 600V

$$m_a = \frac{\sqrt{2} |V_{ac}|}{V_{dc}}$$

$$1 = \frac{\sqrt{2} |V_{ac}|}{V_{dc}} \Rightarrow |V_{ac}| = \frac{1}{\sqrt{2}} \cdot 600 = 424,26$$

$$M_{ax} AC = 424,26 V \text{ (RMS)}$$

$$b) m_c = \frac{2\sqrt{2} |V_{LL}|}{\sqrt{3} V_{dc}} \Rightarrow |V_{LL}| = \frac{\sqrt{3} V_{oc} m_a}{2\sqrt{2}}$$

$$V_{ph} = \frac{V_{LL}}{\sqrt{3}} = \frac{V_{oc} m_a}{2\sqrt{2}} = \frac{600}{2\sqrt{2}} = 212,13 V$$

c) Find AC RMS voltage first:

$$|V_{ac}| = \frac{1}{\sqrt{2}} \cdot 48V \cdot 0,9 = 30,547 V \text{ (RMS)}$$

$$\cos(\varphi_{pf}) = 1 \Rightarrow \text{No reactive power}$$

$$P = VI \Rightarrow I = \frac{P}{V} = \frac{500W}{30,547V} = 16,37 A$$

$$d) P = V_{ac} \cdot I_{ac} \cdot \cos(\varphi_{pf}) \Rightarrow I_{ac} = \frac{P}{V_{ac} \cdot \cos(\varphi_{pf})} = \frac{500W}{30,54V \cdot 0,9} = 18,19 A$$

e) DC power can only be active power.

∴ DC voltage ...

e) DC power can only be active power.  
In DC voltage and current is constant, so they cannot have different phase.

$$f) m_a = \frac{\sqrt{2} |V_{ac}|}{V_{dc}} = \frac{\sqrt{2} \cdot 230V}{400V} = 0,81$$

$$P = V_{ac} \cdot I_{ac} \cdot \cos(\varphi_{pf}) = 230 \cdot 10 \cdot 0,9 = 2070W$$

$$P = V_{ac} \cdot I_{ac} \Rightarrow I_{ac} = \frac{P}{V_{ac}} = \frac{2070}{230} = 9,0A$$

Because power in = power out

Problem 2:

a) A constant  $V_{pv}$  means you won't get max power because the P-V curve changes based on the time of day.

$$b) 1 PM: m_a = \frac{\sqrt{2} |V_{ac}|}{V_{ac}} = \frac{\sqrt{2} \cdot 230}{414,5} = 0,785$$

$$4 PM: \frac{\sqrt{2} \cdot 230}{387} = 0,84$$

$$c) P = V I \cdot \cos \varphi \Rightarrow I_{ac} = \frac{P}{V_{grid}} = \frac{3845}{230} = 16,72A_{RMS}$$

$$V_{ac} = V_{grid} + j X_{grid} \cdot I_{ac} = 230 + 50,16j$$

$$|V_{ac}| = 235,44V_{RMS}$$

$$\angle V_{ac} = \tan^{-1}\left(\frac{50,16}{230}\right) = 12,37^\circ$$

$$V_{ac} = 235,44 \angle 12,37^\circ$$

$$m_a = \frac{\sqrt{2} \cdot 235,44}{414,5} = 0,803$$

d) Same approach as c:

$$I_{ac} = 17,86 \text{ A RMS}$$

$$V_{ac} = 215 + j53,58$$

$$|V_{ac}| = 221,5 \text{ V RMS}$$

$$m_a = \frac{\sqrt{2} \cdot 221,5}{414,5} = 0,756$$

### Problem 3

$$a) n_{\text{mec, gen}} = \frac{120 \cdot f}{P} = \frac{120 \cdot 50}{8} = 750 \text{ rpm}$$

$$\text{Ratio} : \frac{n_{\text{mec, gen}}}{n_{\text{mec, load}}} = \frac{750}{30} = 25$$

$$\text{Without the gear} : f = \frac{n \cdot P}{120} = \frac{30 \cdot 8}{120} = 2 \text{ Hz}$$

Grid needs 50 Hz so 2 Hz is too low.

$$b) \frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{3 \text{ kV}}{33 \text{ kV}} \Rightarrow 1:11 \text{ ratio}$$

$$c) P = T \omega \quad \text{and} \quad \omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 750}{60} = 78,54 \text{ rad/s}$$

$$T_{\text{gen}} = \frac{P}{\omega} = \frac{3 \cdot 10^6}{78,54} = 38209 \text{ Nm}$$

d) 3 phase formula :

$$P = \sqrt{3} V_{LL} \cdot I_L \cdot \cos(\varphi_{pf})$$

$$\Rightarrow I_{\text{grid}} = \frac{P}{\sqrt{3} V_{\text{grid}} \cdot \cos(\varphi_{pf})} = \frac{3 \cdot 10^6}{\sqrt{3} \cdot 33 \cdot 10^3} = \underline{\underline{52,45 \text{ A RMS}}}$$

e) Power flows into turbine

$$n_{\text{mec, load}} = \frac{750}{25} = 30 \text{ rpm}$$

$$f) f_{\text{gen}} = \frac{nP}{120} = \frac{30 \cdot 184}{120} = \frac{5520}{120} = 46 \text{ Hz}$$

$$f) f_{gen} = \frac{117}{120} = \frac{20 \cdot 10^3}{120} = \frac{1}{120} = 46 \text{ Hz}$$

$$g) P = T\omega \Rightarrow T = \frac{P}{\omega}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 30}{60} = \pi$$

$$T = \frac{P}{\omega} = \frac{3 \cdot 10^6}{\pi} = 955 \text{ kNm}$$

h) Advantages:

- Maximum power point tracking
- No gear
- Reduced mechanical stress
- Inverter can match the grid